

Scale-free regret for self-confidence rewards and a corrected combination rule

Proof note for the paper accompanying this repository

Abstract

The paper’s Theorem 1 lower-bounds correctness regret by a constant times $\ln T$. That constant is proportional to $1/s^2$, with $s = (1-\rho)(c_1 - c'_0)$. The factor $(c_1 - c'_0)$ is a numeric scale of the confidence encoding. This note proves a scale-free replacement: any learner whose observation is a garbling of the honest-wrong event pays

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \geq \sum_{a \neq a^*} \frac{\Delta_a}{\text{kl}((1 - \mu_a)(1 - \rho), (1 - \mu_{a^*})(1 - \rho))},$$

which is of order $\ln T/(1 - \rho)$ as $\rho \rightarrow 1$. The power 2 in the paper’s $1/s^2$ bound is proved here only under an extra non-vanishing Bernoulli-variance assumption; it is not true for every encoding of the same latent event. The paper’s Proposition 3 is restated with an explicit hypothesis on the observation mask. Under that hypothesis the per-mask statement holds. The mask-averaged slope need not be positive when the mask depends on correctness, and arm-dependent grounding coverage can reverse the arm ordering. An inverse-propensity correction restores the affine reward when observation propensities are known and missingness is independent of correctness given the arm.

1 Setup

Fix a question type and a finite set of arms (retrieval strategies) $\{1, \dots, K\}$, $K \geq 2$. Arm a returns a correct answer with probability $\mu_a \in (0, 1)$, written $Y \sim \text{Bernoulli}(\mu_a)$. Let a^* be the unique maximizer of μ_a and write $\Delta_a = \mu_{a^*} - \mu_a$. Correctness regret of a (possibly randomized) policy is the random variable

$$\text{Reg}_Y(T) = \sum_{t=1}^T \Delta_{A_t},$$

and statements below refer to its expectation. This is the paper’s correctness regret: it charges the gap in μ , not the gap in the scalar the router treats as a reward.

A reward signal takes values in $[0, 1] \cup \{\perp\}$. The symbol $\perp$ means “not observed”; a skipped round does not update the router.

Assumption 1 (Paper, Assumption 1). A reward R depends on the arm only through correctness:

$$\mathbb{E}[R \mid Y, a, R \neq \perp] = \mathbb{E}[R \mid Y, R \neq \perp].$$

Lemma 2 (Paper, Lemma 1). *Under Assumption 1, with $c_y = \mathbb{E}[R \mid Y = y, R \neq \perp]$ and $s = c_1 - c_0$,*

$$r_a := \mathbb{E}[R \mid a, R \neq \perp] = c_0 + s \mu_a,$$

whenever $\mathbb{P}(R \neq \perp \mid a) > 0$. If this probability does not depend on Y given a , the same identity holds for the unconditional mean of the observed samples.

Proof. $\mathbb{E}[R \mid a, R \neq \perp] = \sum_{y \in \{0,1\}} c_y \mathbb{P}(Y = y \mid a, R \neq \perp)$. If $\mathbb{P}(R \neq \perp \mid Y, a) = \mathbb{P}(R \neq \perp \mid a)$, then $\mathbb{P}(Y = 1 \mid a, R \neq \perp) = \mu_a$, and the display follows. The paper’s one-line proof is the case in which $R \neq \perp$ almost surely, where the conditioning on Y drops out of c_y by Assumption 1 alone. $\square$

The extra clause on $\mathbb{P}(R \neq \perp \mid Y, a)$ is not in the paper’s statement of Lemma 1. It is the missing-data hypothesis already used in the paper’s Proposition 2 when unobserved rounds are skipped, and it is needed as soon as $\perp$ is allowed inside Lemma 1.

2 Honest-wrong event

Definition 3 (Overconfidence ρ). Let P_1 and P_0 be probability measures on a measurable signal space, and let $\rho \in [0, 1]$. The *honest-wrong bit* $W \in \{0, 1\}$ and the self-signal C are drawn as follows, given arm a :

$$\begin{aligned} \mathbb{P}(W = 1 \mid Y = 1, a) &= 0, & \mathbb{P}(W = 1 \mid Y = 0, a) &= 1 - \rho, \\ \mathcal{L}(C \mid W = w, a) &= P_w. \end{aligned}$$

Thus $\mathcal{L}(C \mid Y = 1, a) = P_1$ and $\mathcal{L}(C \mid Y = 0, a) = \rho P_1 + (1 - \rho)P_0$. The bit $W = 1$ is the event “the answer is wrong and the confidence was drawn from the honest-wrong law P_0 rather than copied from the correct-answer law P_1 ”.

Definition 3 is the paper’s overconfidence model with the mixture written on the whole law rather than only on the mean. If R is a $[0, 1]$ -valued encoding of C with $\mathbb{E}_{P_1}[R] = c_1$ and $\mathbb{E}_{P_0}[R] = c'_0$, the paper’s slope formula is the special case

$$c_0 = \rho c_1 + (1 - \rho)c'_0, \quad s_{\text{self}} = (1 - \rho)(c_1 - c'_0).$$

The quantity ρ is a probability, not a correlation.

Assumption 4 (Arm enters only through W). The learner observes a random element X such that $\mathcal{L}(X \mid W, a) = \mathcal{L}(X \mid W)$. In particular this holds for $X = C$, for $X = W$, and for $X = \phi_a(C)$ whenever ϕ_a is a measurable map that may depend on the pulled arm but the conditional law of C given W does not.

Assumption 4 is the scale-free form of “the self-signal’s conditional law given correctness depends on the arm only through the honest-wrong event.” Global rescaling is the case of a single map ϕ . Per-arm rescaling is the case of a family (ϕ_a) . Internal normalization computed by the policy from past data is not a change of X ; the lower bound below applies to every policy, so it already covers that normalization.

Write $\alpha_a = (1 - \mu_a)(1 - \rho) = \mathbb{P}(W = 1 \mid a)$ and $\alpha_{a^*} = (1 - \mu_{a^*})(1 - \rho)$. For $\mu_a \in (0, 1)$ and $\rho \in [0, 1)$ one has $\alpha_a \in (0, 1)$. Bernoulli divergence is

$$\text{kl}(p \parallel q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q},$$

with the convention $0 \ln 0 = 0$.

3 Scale-free lower bound

The model class $\mathcal{M}$ consists of every correctness vector $(\mu_a) \in (0, 1)^K$, with ρ and the conditional law of X given W held fixed. A policy is *strongly consistent* on $\mathcal{M}$ if, for every environment in $\mathcal{M}$ and every $b > 0$,

$$\mathbb{E}[\text{Reg}_Y(T)] = o(T^b).$$

This is the uniformly fast convergence hypothesis in Lai and Robbins [1]. The paper’s Theorem 1 uses the same hypothesis, specialized to Bernoulli observations (“consistent on the Bernoulli family”).

Lemma 5 (Data processing through W). *Under Definition 3 and Assumption 4, for any two arms parameters μ and μ' in $(0, 1)$ with induced honest-wrong probabilities α and α' ,*

$$\text{KL}(\mathcal{L}_\mu(X) \parallel \mathcal{L}_{\mu'}(X)) \leq \text{kl}(\alpha \parallel \alpha').$$

The right-hand side does not depend on (P_0, P_1) or on any encoding of C .

Proof. The joint law of (W, X) factors as $\text{Bernoulli}(\alpha) \otimes K(dx \mid w)$ with a kernel K free of the arm, by Assumption 4. The chain rule gives

$$\begin{aligned} \text{KL}(\mathcal{L}_\mu(W, X) \parallel \mathcal{L}_{\mu'}(W, X)) &= \text{kl}(\alpha \parallel \alpha') \\ &\quad + \mathbb{E}_\mu[\text{KL}(K(\cdot \mid W) \parallel K(\cdot \mid W))] \\ &= \text{kl}(\alpha \parallel \alpha'). \end{aligned}$$

X is a coordinate of (W, X) , so the data-processing inequality yields the claim. $\square$

Lemma 6 (History change of measure). *Let ν and ν' be two environments in $\mathcal{M}$ that differ only in the correctness of a single arm a , and let $\kappa = \text{KL}(\mathcal{L}_\nu(X \mid A = a) \parallel \mathcal{L}_{\nu'}(X \mid A = a))$. For any policy and any horizon T ,*

$$\text{kl}\left(\frac{\mathbb{E}_\nu[N_a(T)]}{T}, \frac{\mathbb{E}_{\nu'}[N_a(T)]}{T}\right) \leq \mathbb{E}_\nu[N_a(T)] \kappa,$$

where $N_a(T) = \sum_{t=1}^T \mathbf{1}\{A_t = a\}$, provided the two arguments of kl lie in $(0, 1)$. If either argument equals 0 or 1, the same bound holds by continuity on $[0, 1]$ whenever the left side is finite, and the history measures are singular whenever the left side is infinite.

Proof. Let $H_T = (A_1, X_1, \dots, A_T, X_T)$ and write P, P' for its laws under ν, ν' . The policy kernel of A_t given the past is the same in both environments, so its conditional divergence is zero. The only nonzero one-step divergence is the divergence of X_t given $A_t = a$. The chain rule therefore gives $\text{KL}(P\|P') = \mathbb{E}_\nu[N_a(T)] \kappa$.

Let $Z = N_a(T)/T \in [0, 1]$ and draw $B \mid Z \sim \text{Bernoulli}(Z)$. Then B is marginally $\text{Bernoulli}(\mathbb{E}[Z])$ under each environment, and B is a randomized function of H_T . Data processing gives

$$\text{kl}(\mathbb{E}_\nu[Z], \mathbb{E}_{\nu'}[Z]) = \text{KL}(\mathcal{L}(B)\|\mathcal{L}'(B)) \leq \text{KL}(P\|P').$$

□

Lemma 6 is the change-of-measure step in the form used by Lai and Robbins. Bretagnolle–Huber [2] is the sibling inequality that transports a single event rather than the mean of $N_a(T)/T$: for every event A ,

$$P(A) + P'(A^c) \geq \frac{1}{2} \exp(-\text{KL}(P\|P')).$$

The proofs below do not rely on that form. They use Lemma 6, whose proof is the preceding paragraph, together with the elementary bound $\text{KL} \leq \chi^2$ recorded in Lemma 7.

Lemma 7 ($\text{KL} \leq \chi^2$ for a binary experiment). *For $p \in [0, 1]$ and $q \in (0, 1)$, $\text{kl}(p\|q) \leq (p - q)^2/(q(1 - q))$.*

Proof. Let $r = \text{dBernoulli}(p)/\text{dBernoulli}(q)$, which exists because $q \in (0, 1)$. Then $\text{kl}(p\|q) = \mathbb{E}_q[r \ln r]$ and $\chi^2 = (p - q)^2/(q(1 - q)) = \mathbb{E}_q[(r - 1)^2]$. The inequality $\ln r \leq r - 1$ for $r > 0$ gives $r \ln r \leq r^2 - r$, and taking $\mathbb{E}_q$ produces $\text{kl}(p\|q) \leq \mathbb{E}_q[r^2] - 1 = \chi^2$. □

Theorem 8 (Scale-free form of the paper’s Theorem 1). *Assume Definition 3 and Assumption 4, and assume $\rho \in [0, 1]$. Let the policy be strongly consistent on $\mathcal{M}$. Then*

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \geq \sum_{a \neq a^*} \frac{\Delta_a}{\text{kl}(\alpha_a \|\alpha_{a^*})}, \quad \alpha_b = (1 - \mu_b)(1 - \rho). \quad (1)$$

The right-hand side depends on the self-signal only through ρ and (μ_b) . It does not depend on (P_0, P_1) , on (c_1, c'_0) , or on any per-arm or global measurable re-encoding of C .

Proof. Fix a suboptimal arm a and an alternative correctness $\mu'_a \in (\mu_{a^*}, 1)$. Let ν be the original environment and let ν' agree with ν except that arm a has correctness μ'_a . In ν' the unique optimal arm is a , with gap $\Delta' = \mu'_a - \mu_{a^*} > 0$ against every other arm. Strong consistency on ν' gives, for every $\varepsilon > 0$ and all large T ,

$$\mathbb{E}_{\nu'}[T - N_a(T)] \leq \frac{\mathbb{E}_{\nu'}[\text{Reg}_Y(T)]}{\Delta'} \leq T^\varepsilon,$$

and strong consistency on ν gives $\mathbb{E}_\nu[N_a(T)] \leq T^\varepsilon$. Set $p_T = \mathbb{E}_\nu[N_a(T)]/T$ and $q_T = \mathbb{E}_{\nu'}[N_a(T)]/T$, so $p_T \leq T^{\varepsilon-1}$ and $q_T \geq 1 - T^{\varepsilon-1}$.

Lemma 5 bounds the one-step divergence of X by $\text{kl}(\alpha_a \parallel \alpha'_a)$, where $\alpha'_a = (1 - \mu'_a)(1 - \rho)$. Lemma 6 therefore yields

$$\text{kl}(p_T \parallel q_T) \leq \mathbb{E}_\nu[N_a(T)] \text{kl}(\alpha_a \parallel \alpha'_a). \quad (2)$$

For the binary divergence, using $0 \ln 0 = 0$,

$$\text{kl}(p_T \parallel q_T) = p_T \ln \frac{p_T}{q_T} + (1 - p_T) \ln(1 - p_T) - (1 - p_T) \ln(1 - q_T).$$

The hypothesis $q_T \geq 1 - T^{\varepsilon-1}$ gives $-\ln(1 - q_T) \geq (1 - \varepsilon) \ln T$. Also $p_T \rightarrow 0$, so $(1 - p_T) \rightarrow 1$, the elementary limit $u \ln u \rightarrow 0$ as $u \rightarrow 0$ gives $p_T \ln(p_T/q_T) \rightarrow 0$, and $(1 - p_T) \ln(1 - p_T) \rightarrow 0$. Hence

$$\text{kl}(p_T \parallel q_T) \geq (1 - \varepsilon) \ln T - o(\ln T).$$

Combine with (2). For every fixed alternative μ'_a and every $\varepsilon > 0$,

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}_\nu[N_a(T)]}{\ln T} \geq \frac{1 - \varepsilon}{\text{kl}(\alpha_a \parallel \alpha'_a)}.$$

The right-hand side may be sent to $1/\text{kl}(\alpha_a \parallel \alpha'_a)$ by letting $\varepsilon \downarrow 0$. This holds for every $\mu'_a \in (\mu_{a^*}, 1)$. Continuity of $\text{kl}(\alpha_a \parallel \cdot)$ on $(0, 1)$ gives $\text{kl}(\alpha_a \parallel \alpha'_a) \downarrow \text{kl}(\alpha_a \parallel \alpha_{a^*})$ as $\mu'_a \downarrow \mu_{a^*}$, so the liminf is at least every number strictly below $1/\text{kl}(\alpha_a \parallel \alpha_{a^*})$, and therefore

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}_\nu[N_a(T)]}{\ln T} \geq \frac{1}{\text{kl}(\alpha_a \parallel \alpha_{a^*})}.$$

Summing $\Delta_a \mathbb{E}[N_a(T)]$ over suboptimal arms gives (1).

Re-encoding does not appear in the argument. If the learner observes $\phi_a(C)$ instead of C , Assumption 4 still holds, Lemma 5 still caps every one-step divergence by the same $\text{kl}(\alpha_a \parallel \alpha'_a)$, and the displayed constant is unchanged. A map ϕ_a can destroy information and make the true divergence smaller, which can only increase the liminf. It cannot push the liminf below (1). $\square$

Corollary 9 (Explicit order $\ln T/(1 - \rho)$). *Under the hypotheses of Theorem 8,*

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \geq \sum_{a \neq a^*} \frac{\mu_{a^*}(1 - \mu_{a^*})}{(1 - \rho) \Delta_a}. \quad (3)$$

In particular, for any fixed (μ_a) with a unique optimum and $\mu_{a^} \in (0, 1)$, the right-hand side of (1) is $\Theta(1/(1 - \rho))$ as $\rho \rightarrow 1^-$.*

Proof. Apply Lemma 7 with $p = \alpha_a$ and $q = \alpha_{a^*}$:

$$\text{kl}(\alpha_a \| \alpha_{a^*}) \leq \frac{(\alpha_a - \alpha_{a^*})^2}{\alpha_{a^*}(1 - \alpha_{a^*})} = \frac{(1 - \rho) \Delta_a^2}{(1 - \mu_{a^*})(1 - (1 - \mu_{a^*})(1 - \rho))}.$$

The identity $1 - (1 - \mu_{a^*})(1 - \rho) = \mu_{a^*} + \rho(1 - \mu_{a^*}) \geq \mu_{a^*}$ gives

$$\text{kl}(\alpha_a \| \alpha_{a^*}) \leq \frac{(1 - \rho) \Delta_a^2}{\mu_{a^*}(1 - \mu_{a^*})},$$

and substituting into (1) gives (3).

For the limit, set $t = 1 - \rho$, $\lambda = \Delta_a / (1 - \mu_{a^*})$, and $\alpha_{a^*} = (1 - \mu_{a^*})t$, so $\alpha_a = \alpha_{a^*}(1 + \lambda)$. Then

$$\begin{aligned} \frac{\text{kl}(\alpha_a \| \alpha_{a^*})}{t} &= (1 - \mu_{a^*})(1 + \lambda) \ln(1 + \lambda) \\ &\quad + \frac{1 - \alpha_{a^*}(1 + \lambda)}{t} \ln \frac{1 - \alpha_{a^*}(1 + \lambda)}{1 - \alpha_{a^*}}. \end{aligned}$$

As $t \rightarrow 0$, $\ln((1 - u)/(1 - v)) = (v - u) + O(t^2)$ with $u = \alpha_{a^*}(1 + \lambda)$ and $v = \alpha_{a^*}$, and the second term tends to $-(1 - \mu_{a^*})\lambda$. Therefore

$$\lim_{\rho \rightarrow 1^-} \frac{\text{kl}(\alpha_a \| \alpha_{a^*})}{1 - \rho} = (1 - \mu_{a^*})[(1 + \lambda) \ln(1 + \lambda) - \lambda], \quad (4)$$

which is strictly positive for $\lambda > 0$. Combined with the χ^2 upper bound, $\text{kl}(\alpha_a \| \alpha_{a^*}) = \Theta(1 - \rho)$. $\square$

Proposition 10 (Where the paper's power 2 holds). *Suppose that, in addition to the hypotheses of Theorem 8, the learner observes only the Bernoulli trick $B \sim \text{Bernoulli}(R)$ of a reward $R \in [0, 1]$ whose means $r_a = \mathbb{E}[R \mid a]$ satisfy $r_{a^*} - r_a = s\Delta_a$ with*

$$0 < s \leq (1 - \rho)(c_1 - c'_0)$$

and $r_{a^} \in [\delta, 1 - \delta]$ for some $\delta \in (0, 1/2]$ independent of ρ . Then*

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \geq \sum_{a \neq a^*} \frac{\delta(1 - \delta)}{(1 - \rho)^2 (c_1 - c'_0)^2 \Delta_a}. \quad (5)$$

If R itself is i.i.d. Bernoulli(r_a), the same argument with Lemma 7 reproduces the paper's constant:

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \geq \sum_{a \neq a^*} \frac{r_{a^*}(1 - r_{a^*})}{s^2 \Delta_a}.$$

Proof. The one-step law of B is Bernoulli(r_a), so Lemma 6 and the argument of Theorem 8 give

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[N_a(T)]}{\ln T} \geq \frac{1}{\text{kl}(r_a \| r_{a^*})}.$$

Lemma 7 yields $\text{kl}(r_a \| r_{a^*}) \leq (r_{a^*} - r_a)^2 / (r_{a^*}(1 - r_{a^*})) \leq s^2 \Delta_a^2 / (\delta(1 - \delta))$. Insert $s \leq (1 - \rho)(c_1 - c'_0)$ and multiply by Δ_a . The second display is the same calculation with the exact factor $r_{a^*}(1 - r_{a^*})$ left in place, which is the paper's Theorem 1. $\square$

Remark 11 (Which power of $(1 - \rho)$ is actually proved). Three statements have different hypotheses, and only the first is scale-free.

1. Theorem 8 and Corollary 9 prove the power 1: order $\ln T / (1 - \rho)$, with a constant free of (c_1, c'_0) . The channel assumption is Definition 3 plus Assumption 4. This is the bound that survives an arbitrary measurable re-encoding.
2. Proposition 10 proves the power 2, of order $\ln T / (1 - \rho)^2$, only when the observation is a Bernoulli trick (or an exact Bernoulli reward) whose mean stays in a ρ -independent subinterval of $(0, 1)$ and whose mean gap is at most order $(1 - \rho)(c_1 - c'_0)$. The constant depends on the scale $(c_1 - c'_0)$ and on the variance floor δ . Shrinking all confidence values toward $1/2$ shrinks $c_1 - c'_0$ and inflates this particular lower bound; that inflation tracks a real loss of Bernoulli-trick information, not a defect in the χ^2 step. Expanding the encoding inside $[0, 1]$ can increase $c_1 - c'_0$ up to 1, but it cannot drive the constant in Theorem 8 below the honest-wrong divergence.
3. The universal claim “every self-confidence encoding pays $\Omega(\ln T / (1 - \rho)^2)$ ” is false. If $X = W$, then $s = 1 - \rho$ and $r_{a^*}(1 - r_{a^*}) = \alpha_{a^*}(1 - \alpha_{a^*}) = \Theta(1 - \rho)$, so the paper's own formula $r(1 - r) / (s^2 \Delta)$ is already $\Theta(1 / (1 - \rho))$, in agreement with (4). The variance floor δ independent of ρ fails for this encoding, which is why Proposition 10 does not apply.

A matching upper bound of order $\ln T / (1 - \rho)$ holds on the channel $X = W$. For each fixed $\rho \in [0, 1)$, the honest-wrong bits are an ordinary Bernoulli bandit with means $\alpha_a \in (0, 1)$. KL-UCB [3] satisfies

$$\limsup_{T \rightarrow \infty} \frac{\mathbb{E}[N_a(T)]}{\ln T} \leq \frac{1}{\text{kl}(\alpha_a \| \alpha_{a^*})}$$

on every such instance. Each pull of arm a costs Δ_a in correctness regret, so

$$\limsup_{T \rightarrow \infty} \frac{\mathbb{E}[\text{Reg}_Y(T)]}{\ln T} \leq \sum_{a \neq a^*} \frac{\Delta_a}{\text{kl}(\alpha_a \| \alpha_{a^*})}.$$

Together with (4), the leading power of $(1 - \rho)$ on this channel is exactly 1. The square-power expression is a valid lower bound for encodings that keep a ρ -independent Bernoulli variance; it is not a lower bound for every function of the honest-wrong event.

Proposition 12 (No information when $\rho = 1$). *If $\rho = 1$, then $\mathcal{L}(X \mid a)$ does not depend on a . For any policy,*

$$\frac{1}{K!} \sum_{\sigma} \mathbb{E}_{\sigma}[\text{Reg}_Y(T)] = \frac{T}{K} \sum_a \Delta_a,$$

where the average is over uniform random relabelings of a fixed correctness vector. In particular some arm labeling forces $\mathbb{E}[\text{Reg}_Y(T)] = \Theta(T)$ whenever $\sum_a \Delta_a > 0$.

Proof. $\rho = 1$ forces $W = 0$ almost surely, so Assumption 4 makes X independent of the arm. The law of the action sequence is therefore the same under every relabeling. Averaging the nonnegative gaps gives the identity. $\square$

Conjecture 13 (Fractional Beta update). Let $R \in [0, 1]$ be a non-degenerate encoding of W , so that $\mathbb{P}(R \in (0, 1)) > 0$, and let the router apply the fractional update $\alpha \leftarrow \alpha + R$, $\beta \leftarrow \beta + (1 - R)$. For each fixed $\rho \in [0, 1)$, this policy attains

$$\mathbb{E}[\text{Reg}_Y(T)] = O\left(\frac{\ln T}{1 - \rho}\right)$$

as $T \rightarrow \infty$, with a constant depending on (μ_a) and on the law of R given W , and not growing faster than $1/(1 - \rho)$.

When the reward is already the bit $W \in \{0, 1\}$, the fractional update coincides with the integer Beta update, and the $O(\ln T)$ bound of Agrawal and Goyal [4] applies directly; dividing the honest-wrong gap by $1 - \rho$ again yields order $\ln T/(1 - \rho)$. The open case is a reward that takes values in $(0, 1)$. Their proof draws $B \sim \text{Bernoulli}(R)$ first. The conditional expectation of B equals R , but the posterior variance and the boundary terms in that proof are statements about B , and equality of one-step means does not transfer them to $\alpha \leftarrow \alpha + R$. The paper records that the two updates are close in simulation; that is evidence, not a proof. Theorem 8 already lower-bounds the fractional update whenever the resulting policy is strongly consistent, because the lower bound does not depend on how the posterior is written.

4 Combination and the observation mask

Let signals $i = 1, \dots, m$ have fixed weights $w_i > 0$ and potential values $V_i \in [0, 1]$. The learner sees a mask $S \subseteq \{1, \dots, m\}$ and the values $\{V_i : i \in S\}$. The combined reward on $\{S = s\}$ for $s \neq \emptyset$ is

$$R = \frac{\sum_{i \in s} w_i V_i}{\sum_{i \in s} w_i}, \tag{6}$$

and $R = \perp$ on $\{S = \emptyset\}$, in which case the router is not updated. This is the paper's combination rule.

Assumption 14 (Mask-conditional form of Assumption 1). For every signal i , every mask $s \ni i$, and every $y \in \{0, 1\}$,

$$\mathbb{E}[V_i \mid Y = y, a, S = s] = c_{i,y}$$

does not depend on a or on s . Write $s_i = c_{i,1} - c_{i,0}$.

Assumption 15 (Mask law). The mask is conditionally independent of the arm given correctness: $S \perp A_{\text{env}} \mid Y$, where A_{env} denotes the arm that generated the answer. Equivalently, $\mathbb{P}(S = s \mid Y, a) = \mathbb{P}(S = s \mid Y)$.

Assumption 15 is the hypothesis the paper's proof of Proposition 3 uses when it conditions on the observed set and then drops the arm. It is not implied by Assumption 1 for the individual signals. The symbol A_{env} is the arm, not the action random variable; the two coincide on the round being scored.

Proposition 16 (Corrected form of the paper's Proposition 3). *Assume Assumptions 14 and 15, and assume $s_i > 0$ for every i .*

1. *For every nonempty mask s ,*

$$\mathbb{E}[R \mid Y = y, a, S = s] = \frac{\sum_{i \in s} w_i c_{i,y}}{\sum_{i \in s} w_i},$$

which does not depend on a , and the mask-conditional slope

$$s(s) = \frac{\sum_{i \in s} w_i s_i}{\sum_{i \in s} w_i}$$

is strictly positive.

2. $\mathbb{E}[R \mid Y = y, a, R \neq \perp]$ *does not depend on a . The combined reward therefore satisfies Assumption 1.*
3. *If, in addition, $S \perp Y$, then the marginal slope on $\{R \neq \perp\}$ equals $\sum_{s \neq \emptyset} s(s) \mathbb{P}(S = s \mid S \neq \emptyset) > 0$, and Lemma 2 gives $r_a = c_0 + \bar{s} \mu_a$ with that slope $\bar{s}$. The reward-optimal arm is a^* .*

Proof. Part 1 is linearity of conditional expectation under Assumption 14, and $s(s)$ is a convex combination of the s_i . For part 2,

$$\mathbb{E}[R \mid Y = y, a, S \neq \emptyset] = \sum_{s \neq \emptyset} \mathbb{E}[R \mid Y = y, S = s] \mathbb{P}(S = s \mid Y = y, a, S \neq \emptyset).$$

Assumption 15 removes a from the mask probabilities, and part 1 removes a from the summands.

For part 3, $S \perp Y$ makes $\mathbb{P}(S = s \mid Y = y, S \neq \emptyset)$ independent of y , so

$$c_1 - c_0 = \sum_{s \neq \emptyset} s(s) \mathbb{P}(S = s \mid S \neq \emptyset).$$

Each $s(s) > 0$ and at least one mask is charged by $\mathbb{P}(S \neq \emptyset)$, which is positive on the event we condition on. Lemma 2 needs $\mathbb{P}(R \neq \perp \mid Y, a)$ independent of Y . Here $R \neq \perp$ is exactly $S \neq \emptyset$, and $S \perp Y$ together with Assumption 15 makes this probability a constant. The resulting r_a is strictly increasing in μ_a . $\square$

The slope formula written in the paper, $\sum_i w_i s_i / \sum_i w_i$, is the case of part 1 in which s is the full index set, and it is the case of part 3 when a single mask carries all the mass. It is not the marginal slope merely from Assumptions 14 and 15.

Remark 17 (Positive per-mask slopes do not imply a positive marginal slope). Assumption 15 is not enough for $c_1 > c_0$. The marginal slope admits the decomposition

$$c_1 - c_0 = \sum_{s \neq \emptyset} s(s) \mathbb{P}(S = s \mid Y = 1, S \neq \emptyset) + \sum_{s \neq \emptyset} c_0(s) \left(\mathbb{P}(S = s \mid Y = 1, S \neq \emptyset) - \mathbb{P}(S = s \mid Y = 0, S \neq \emptyset) \right),$$

where $c_0(s) = \sum_{i \in s} w_i c_{i,0} / \sum_{i \in s} w_i$. The second sum can dominate.

Example. Two signals, equal weights. Signal A has $(c_1, c_0) = (0.1, 0)$, so $s_A = 0.1$. Signal B has $(c_1, c_0) = (0.95, 0.9)$, so $s_B = 0.05$. Given $Y = 1$, the mask is $\{A\}$ almost surely. Given $Y = 0$, the mask is $\{B\}$ almost surely. Then S is a function of Y , so Assumption 15 holds, and both signals satisfy Assumption 14. The combined reward is observed on every round, with

$$\mathbb{E}[R \mid Y = 1] = 0.1, \quad \mathbb{E}[R \mid Y = 0] = 0.9.$$

The marginal slope equals -0.8 . Hence $r_a = 0.9 - 0.8\mu_a$, which is strictly decreasing in correctness, and the reward-optimal arm is the worst arm. Part 1 of Proposition 16 still holds on each singleton mask: the failure is only in passing from the mask-conditional slopes to the scalar the router actually averages.

Remark 18 (Arm-dependent grounding coverage). Drop Assumption 15, and let the mask depend on the arm even after conditioning on Y . Order can reverse while every individual signal keeps a positive, arm-free slope. The following numbers are a grounding pattern: one strategy almost always produces a claim-extractable answer, so the weak lexical signal is present, while the other strategy almost always misses that signal and is scored by a rare high-slope judge instead. Both propensities lie in $(0, 1)$, which matters for the correction in Proposition 19.

Signals G and J , equal weights, exactly one of them observed, independent of Y given the arm:

arm	μ	$\mathbb{P}(S = \{G\} \mid a)$	$\mathbb{P}(S = \{J\} \mid a)$	conditional laws
a^*	0.9	0.99	0.01	$G: (c_1, c_0) = (0.52, 0.50)$
b	0.7	0.01	0.99	$J: (c_1, c_0) = (1, 0)$

Signal slopes are $s_G = 0.02$ and $s_J = 1$, both positive and arm-free. Direct expansion gives

$$\begin{aligned} \mathbb{E}[R \mid a^*] &= 0.9(0.99 \cdot 0.52 + 0.01 \cdot 1) + 0.1(0.99 \cdot 0.50) = 0.52182, \\ \mathbb{E}[R \mid b] &= 0.7(0.01 \cdot 0.52 + 0.99 \cdot 1) + 0.3(0.01 \cdot 0.50) = 0.69814. \end{aligned}$$

So $r_b > r_{a^*}$ while $\mu_b < \mu_{a^*}$. Each signal alone would have preserved the order: the G -only means are 0.518 and 0.514, and the J -only means are 0.9 and 0.7. The reversal is produced entirely by which signal the arm makes available.

The same construction is what arm-dependent grounding coverage does to a two-signal reward. One strategy extracts claims on almost every round, so grounding is almost always the observed signal. The other strategy rarely extracts a claim, so the remaining signal is the one that enters the average.

Proposition 19 (What survives). 1. Single signal. Let $m = 1$, and let Assumption 14 hold with slope $s > 0$. Let the miss probability be free of Y :

$$\mathbb{P}(1 \notin S \mid Y, a) = \mathbb{P}(1 \notin S \mid a).$$

Skipping unobserved rounds yields

$$\mathbb{E}[V \mid a, S \neq \emptyset] = c_0 + s\mu_a.$$

Arm-dependent coverage changes how often the arm is updated and does not change the mean of the updates. The reward-optimal arm remains a^* .

2. Constant fill. If, instead, an unobserved round is replaced by a constant κ , the learned mean is $(1 - m_a)(c_0 + s\mu_a) + m_a\kappa$ with $m_a = \mathbb{P}(S = \emptyset \mid a)$. This is the paper's Proposition 2. Whenever m_a is strictly larger on the better arm and κ lies below both conditional means, the ordering can reverse. Skipping and filling are different estimators; coverage that is harmless for the first is not harmless for the second.

3. Inverse propensity, several signals. Suppose Assumption 14 holds in the stronger form $\mathbb{E}[V_i \mid Y = y, a] = c_{i,y}$, the mask is independent of $(Y, (V_i)_i)$ given the arm, and the propensities $\pi_a(i) = \mathbb{P}(i \in S \mid a)$ are known and satisfy $\pi_a(i) \geq \pi_{\min} > 0$. The corrected reward

$$\hat{R} = \frac{\sum_{i=1}^m w_i V_i \mathbf{1}\{i \in S\} / \pi_a(i)}{\sum_{i=1}^m w_i} \quad (7)$$

obeys $\mathbb{E}[\hat{R} \mid a] = c_0 + \bar{s} \mu_a$ with $\bar{s} = \sum_i w_i s_i / \sum_i w_i$. If every $s_i > 0$, then $\bar{s} > 0$ and the reward-optimal arm is a^* .

On the numerical instance of Remark 18 this correction returns

$$\mathbb{E}[\hat{R} \mid a^*] = 0.709, \quad \mathbb{E}[\hat{R} \mid b] = 0.607,$$

so the order matches correctness. The propensities used here are signal-observation propensities. The router's logged selection propensity is the probability of the arm under the logging policy; reweighting by that number corrects a different bias and does not estimate $\pi_a(i)$.

Proof. For part 1, $\mathbb{P}(Y = 1 \mid a, S \neq \emptyset) = \mu_a$ because coverage is independent of Y given a , and Assumption 14 supplies the conditional means of V . Part 2 is

the law of total expectation, splitting on the observation indicator; the paper’s Proposition 2 is this identity plus the observation that a large m_a pulls r'_a toward κ .

For part 3, independence of the mask and V_i given a gives

$$\mathbb{E} \left[\frac{V_i \mathbf{1}\{i \in S\}}{\pi_a(i)} \mid a \right] = \mathbb{E}[V_i \mid a] = c_{i,0} + s_i \mu_a.$$

Linearity produces $\mathbb{E}[\hat{R} \mid a]$. The two numerical values are

$$\frac{1}{2}((0.50 + 0.02 \cdot 0.9) + 0.9) = 0.709, \quad \frac{1}{2}((0.50 + 0.02 \cdot 0.7) + 0.7) = 0.607.$$

□

If the mask depends on Y given the arm, part 3 does not apply: the weight that would be correct is $\mathbb{P}(i \in S \mid a, Y)$, and Y is not observed by the router. No claim is made for that propensity. Remark 17 is the boundary case in which the mask is a function of Y , Assumption 15 holds, and the uncorrected marginal slope is already negative.

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